

EMBODIED

Two hours to explore how robots learn and act in the real world, without leaving simulation.



Franka Panda, seven joints, facing the drawer cabinet. Isaac Sim 5.1 path-traced render, NVIDIA L40S.

HOW TO GET IN

At the workshop, each team receives a card with its machine's address, its username and its password.

Afterwards, everything is open source:

github.com/charbelsan/embodied-ai-workshop. Run it on your own NVIDIA RTX machine, host it in the cloud, or try the team challenge on any laptop with the CPU edition.

Open the address in Chrome, Edge or Firefox. Everything runs in your browser, nothing to install.

AT HOME: `./START.SH`
OPENS THE SAME
NOTEBOOK

Enter *the simulation*

Each team has its own GPU machine, accessible in a browser. The robot is already there, ready for you to get started.

THE ADDRESS Open your team's address in Chrome, Edge or Firefox. No installation is needed.

CERTIFICATE WARNING normal on a temporary machine: choose Advanced, then Proceed (Chrome, Edge) or Accept the Risk and Continue (Firefox).

SIGN IN with your team's username and password. The desktop opens with Isaac Sim already running.

MINIMISE ISAAC SIM to uncover the desktop icons. Double-click START WORKSHOP. Checks run, READY appears and the guided notebook opens in the browser.

IF NOT READY APPEARS run START WORKSHOP once more. If it persists, ask a facilitator.

THE NOTEBOOK Read and run cells in order with Shift + Enter. To watch the arm move during a live rollout, return to Isaac Sim.



The kitting workstation of the team challenge, seen from above: parts on the work table, the open drawer, the tray.

ON YOUR DESKTOP

START WORKSHOP	checks the machine and opens the notebook
Isaac Sim	the live robot scene
Jupyter Lab	the guided notebook
VS Code	your team's code, in <code>team/</code>
Terminal	evaluate, train, submit
Results	scores, traces and videos
Challenge sheet	the team challenge instructions
RESTART WORKSHOP	start everything over, deleting nothing



Gripper on the handle. The visual student uses only its two cameras and joint readings to find it.



The motherboard's memory slots. The RAM stick must fit with submillimetre clearance, guided by vision.

“ok is the skill's report. The world state is what matters.”

TWO HOURS

45 minutes of guided experiments: a first motion, reinforcement learning, imitation and DAgger, then generalist models. Next, the team challenge: tidy a kitting station, measure, improve and submit.

A failed motion is often part of the experiment. For a lost connection or an error message, raise your hand and show it to a facilitator.